Physics of the Earth and Planetary Interiors, 7 (1973) 450–460
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IMPLICATIONS FROM FOSSIL MAGNETISM OF THE FEN COMPLEX (CAMBRIAN), SOUTH NORWAY*

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Accepted for publication February 28, 1973

A study of the remanent magnetization of the Fen alkaline/carbonatite rock province of South Norway has revealed a complex magnetic structure. The evolution of this record is thought to be largely associated with an extensive period of metasomatic activity coupled with at least one geomagnetic field inversion. The resulting composite magnetization which may be of fairly high stability has proved difficult to split into its subcomponents and, therefore, palaeomagnetic figures can not be presented. However, the directional trends during demagnetization suggest a major magnetic axis in concordance with other Lower–Middle Palaeozoic results from Western Europe. It is suggested that the Lower–Middle Palaeozoic pole relative to Western Europe remained fixed (at about 160° E 20° N in present grids) for a period of about 300 m.y. For palaeomagnetic reasons some dolerite dikes crossing the Fen complex appear to be of pre–mid–Carboniferous origin though K/Ar analysis indicate mid–Permian ages. It is suggested that a complete reset of the radiometric clock of these dikes took place as a result of the Permian magmatic activity in the closely situated Oslo graben.

1. Introduction

Little information of Lower Palaeozoic palaeomagnetism for Western Europe is presently at hand and the few data made available so far are in part discrepant. Storetvedt (1967) has tried to synthesize the available information, suggesting a fairly fixed pole of rotation relative to Western Europe for a major part of the Lower and Middle Palaeozoic. The purpose of investigating the fossil magnetic record of the Fen complex, which appears to date back to the Early Cambrian, was firstly that of trying to obtain more information on the lower age limit for the postulated Lower–Middle Palaeozoic pole and secondly to collect data for a future study of the polar pattern at around the time of the Eocambrian glaciation.

Deviating Palaeozoic poles have been reported for Ordovician (Deutsch and Somayajulu, 1970) as well as for Silurian formations (Briden et al., 1970; Evans et al., 1971). The authors concerned tend to interpret the discrepant results as largely revealing relatively shortlived periods of instability of the earth' s axis of rotation. Contrasting with this view the present author is inclined to believe that the deviating results are false palaeomagnetic directions, being largely the product of too simplified preconceptions having been applied on complex magnetic structures. The Fen formation provides another example of a complex magnetic build-up where interpretation pitfalls are near at hand.

2. Geological framework and rock collection details

The Fen alkaline/carbonatite rock province is located some 120 kilometres southwest of Oslo (immediately west of the western boundary of the Oslo graben) where it intersects Precambrian gneisses. The geological setting of the complex (in addition to the central vent a large number of surrounding dikes, breccias and plugs occur) is shown in Fig. 1. The most recent geological review of the area is given by Barth and Ramberg (1966).

With his classic paper of 1921 Brögger put forward the hypothesis that alkaline/carbonatite complexes are of magmatic origin. Comprehensive field and laboratory studies in more recent years have strongly supported this idea. On the petrogenesis of the Fen complex, Barth and Ramberg (1966) favour the idea of a liquid magma fraction (giving rise to the

* Norwegian Geotraverse Contribution No. 74.

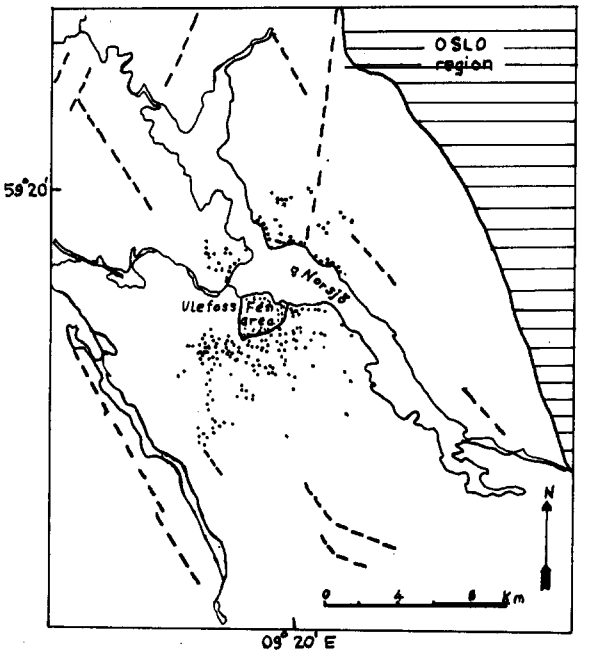


Fig. 1. Sketch map of parts of eastern Norway showing the situation of the Fen volcanic vent (and its associated plugs etc.) relative to the Oslo graben. Broken lines indicate faults or tectonic breccias. Simplified after Barth and Ramberg (1966).

alkaline silicates) separated from a "gas phase" (carbonatite magma) but a time sequence of the different rocks has become hard to establish. Fluidization and gas coring rather than explosive processes may account for the dynamics creating the Fen "volcanic vent" (Barth and Ramberg, 1966). After the major rock types had been emplaced hydrothermal metasomatic changes apparently continued for a long time giving rise to destruction and modification of primary features as well as producing transitional rock types. Furthermore, the Fen vent as well as the host rocks are crossed by steeply dipping dolerite dikes.

A large number of radiometric age determinations on Fen rocks and minerals are available. The most reliable data which have been compiled in Table I suggest a Lower Cambrian/Eocambrian age of the complex which seems to have been subsequently intersected by Permian dolerites. There is no evidence of post-magmatic tectonism of importance for the present study.

Using an ordinary geological orientation method 30 hand samples from 7 sites, comprising dolerite dikes (2 sites), kimberlite (1 site) and haematite carbonatites (4 sites), have been collected. Collection details can be obtained from Table II. The dolerite of site 1 has been subjected to K/Ar analysis (FM Consultants Ltd., 1968). The dating laboratory describes the rock as "an altered, very fine-grained microporphyritic alkali-olivine-dolerite. It does contain a little completely fresh plagioclase, but generally its primary mineralogy is being progressively replaced by secondary chlorite, calcite and sericite". The second dolerite (site 3) which is dated by another laboratory (Priem et al.,1968) is described as "a very fine-grained, dark, completely altered rock, made of calcite-sericite pseudomorphs after felspar in a matrix of calcite, chlorite, epidote and opaques".

TABLE I

Radiometric age determinations from the Fen complex

Rock	Mineral	Method	Age (m.y.)	Reference
Kimberlite	biotite	K/Ar	573	Brock (1964)
Kimberlite	biotite	K/Ar	585	Brock (1964)
Kimberlite	biotite	K/Ar	572 ± 30	Priem et al. (1968)
Kimberlite	biotite	Rb/Sr	533 ± 15	Priem et al. (1968)
Kimberlite	biotite	K/Ar	599 ± 13	FM Consultants (1968)
Hollaite pegmatite	felspar	K/Ar	413	Kley and Schmidlin (1960)
Sövite	biotite	K/Ar	565	Faul et al. (1959)
Sövite	biotite	K/Ar	568	Brock (1964)
Sövite	biotite	K/Ar	590	Brock (1964)
Sövite	biotite	K/Ar	603	Brock (1964)
Dolerite	whole rock	K/Ar	243 ± 7	FM Consultants (1968)
Dolerite	whole rock	K/Ar	250 ± 10	Priem et al. (1968)

TABLE II
Rock collection details

Site no.	Sample nos.	Locality	Rock type
1	Fe 1–4	at Fen iron mine	dolerite dike
2	Fe 5, 6, 29, 30	at Fen iron mine	haematite carbonatite (rödberg)
3	Fe 7–10	in Söve mine	dolerite dike
4	Fe 11–14	Söve farm, at Damtjern	kimberlite
5	Fe 15–19	Vipeto	haematite carbonatite (rödberg)
6	Fe 20–21	at Fen centre	haematite carbonatite (rödberg)
7	Fe 22–28	near Fen iron mine	haematite carbonatite (rödberg)

A sample from the collected kimberlite shows "a highly calcified medium-grained igneous rock rich in biotite phenocrysts. Originally, the matrix may have been composed of olivine, pyroxene or amphibole, mica, magnetite, nepheline and/or alkali-felspars, but these primary minerals have been almost completely replaced by secondary minerals" (FM Consultants Ltd., 1968).

The haematite–carbonatite or rödberg is a metasomatic rock consisting of calcite, ankerite and a considerable portion (20–50%) of haematite (Barth and Ramberg, 1966).

3. High field magnetization versus temperature

Some useful information regarding the magnetic minerals has been achieved by thermomagnetic cycling experiments up to 700°C. The applied fields, giving an approximate saturation of bulk magnetization, varied between 2 kG and 6 kG depending on the amount of stronger magnetic phases. The heating was done in air. In order to obtain a certain indication to what extent oxidation during heating distorts the true $J_S(T)$-characteristics of the natural magnetic mineralogy two different types of specimens have been analysed: finely crushed material and rock chips. Characteristic results are shown in Fig. 2.

Chip specimens of dolerite and of kimberlite behave in a rather identical manner, giving mostly single-phased curves with T_C sharply peaked around 550°C. Dolerite specimens from site 1 may have an additional inflection around 400°C. All the cooling curves appear to be single-phased, but they always lie a little below the heating curves. The results of powdered specimens differ from the chip ones in that the heating and cooling curves diverge more strongly (cf. the results of specimen 7-a2, Fig. 2). This effect appears to be much less pronounced in kimberlite specimens than in the dolerites where the specific magnetization after heating may be reduced by as much as 70%. By reversing the temperature at various stages on the heating cycle it seems that the lowering of the bulk spontaneous magnetization initiates at temperatures below 400°C, though this effect has a major advance above 600°C.

The carbonatite specimens show a diversified magnetic mineralogy. A phase with Curie temperature about 660°C seems nearly invariably present but many of the heating curves have marked inflections at about 350°C as well as around 560°C. On the whole, the 350°C-component contributes about 50% of the original total spontaneous magnetization of this rock. However, the heating experiments show that this latter phase is metastable, dissappearing completely on heating through the temperature range 250°C–400°C (approximately). Consequently, specimens which contain this metastable phase exhibit strongly irreversible $J_S(T)$ behaviour at lower temperatures. For the carbonatite rock there appears to be no difference between the results of powder specimens and of chip specimens.

The thermomagnetic experiments suggest there-

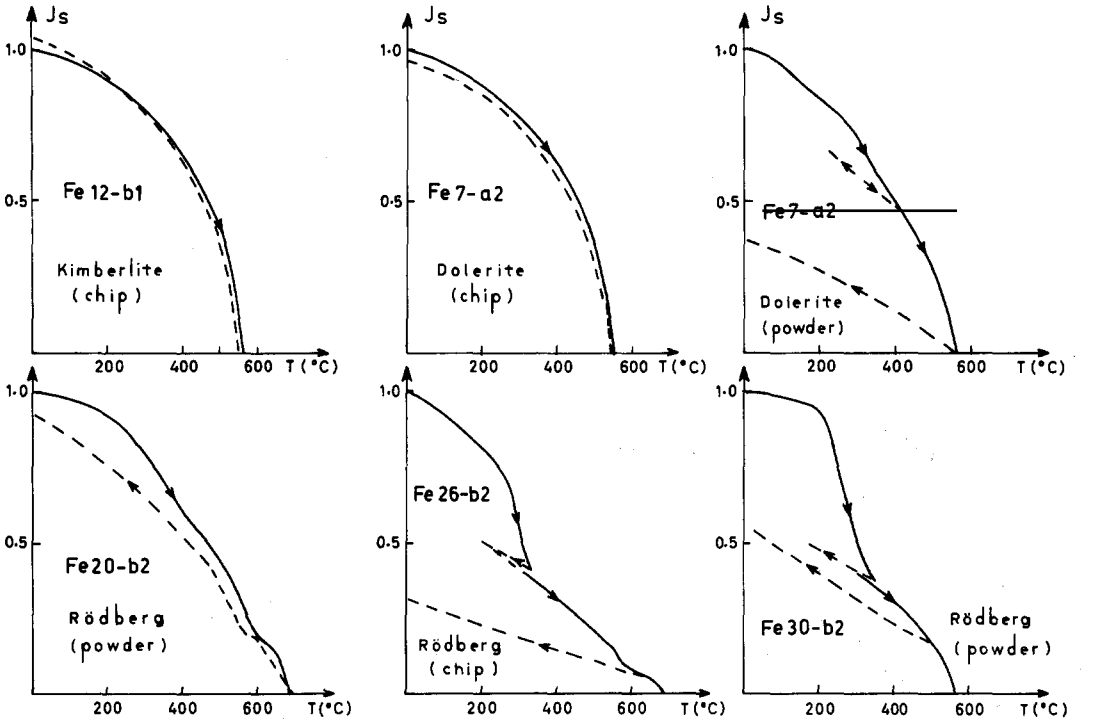


Fig. 2. Representative examples of saturation magnetization (arbitrary scale) versus temperature. Note the difference between a chip specimen and a powdered specimen in sample no. 7 (dolerite).

fore that the haematite carbonatite contain the expected haematite in addition to a certain fraction of magnetite and a metastable mineral which is most likely maghemite (it seems likely that the latter phase inverts into haematite on heating, but the method applied is not sufficiently sensitive to detect a small additional amount of such a weak magnetic mineral). Furthermore, magnetite ($T_c \sim 550°C$) appears to be the predominating magnetic phase in the kimberlite as well as in the dolerites. Maghemite tends to be partly present in the dolerites; a higher stage of "maghemitization" in these rocks compared to the kimberlite may perhaps account for the difference in $J_S(T)$ behaviour on powdered material.

4. Experimental analysis of the magnetic record

With the intention of discriminating between possible sub-components of magnetization, specimens from all collected samples have been subjected to stepwise progressive thermal and/or alternating field demagnetization. For measurement after cooling from each demagnetization step astatic magnetometers have been used. The average N.R.M. intensity for dolerite, kimberlite and carbonatite specimens was found to be approximately $6 \cdot \cdot 10^{-4}$ gauss, $1 \cdot 10^{-4}$ gauss and $5 \cdot 10^{-5}$ gauss, respectively. In an attempt to record events of possible mineralogical changes (as a function of thermal treatment) magnetic susceptibility for a number of specimen has been measured after each heating step.

The natural remanence intensity (J_n) for the dolerites as well as for the kimberlite approaches zero at around 560°C. This figure agrees with the J_S-T data suggesting nearly pure magnetite as the major magnetization carrier of these rocks. For the haematite carbonatite about 50% of analyzed specimens suggest haematite as the only remanence component; the J_S-T curves decaying to zero at around 660°C. In these latter cases the J_n-decay may either be sharply concentrated at temperatures above 600°C or may be linear throughout the whole range of temperatures studied. For the remaining carbonatite specimens it seems that the bulk remanence is associated with a

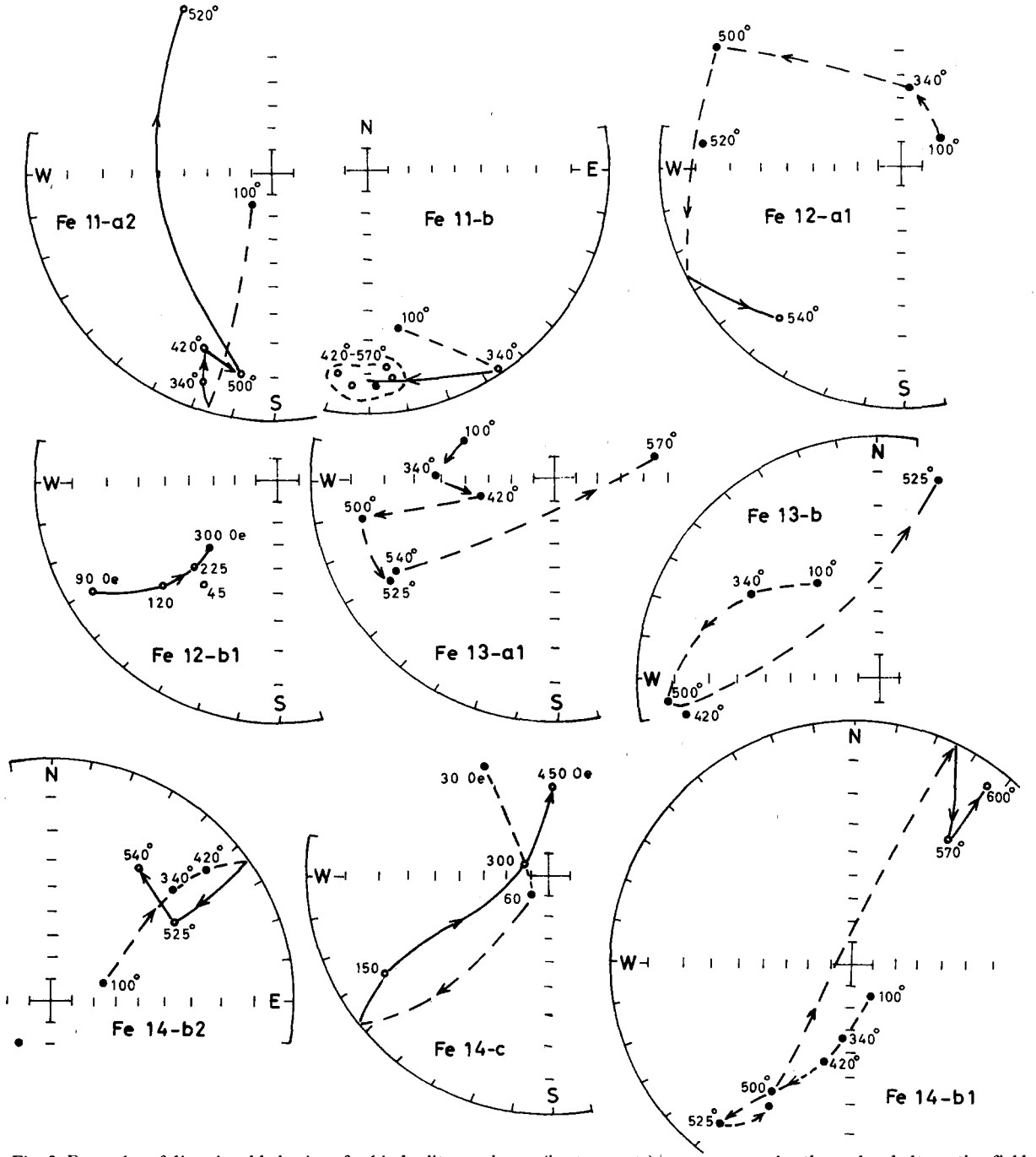


Fig. 3. Examples of directional behaviour for kimberlite specimens (in stereonets) versus progressive thermal and alternating field demagnetization. Full circles represent downward inclinations, open circles are upward seeking magnetism. See text for further details.

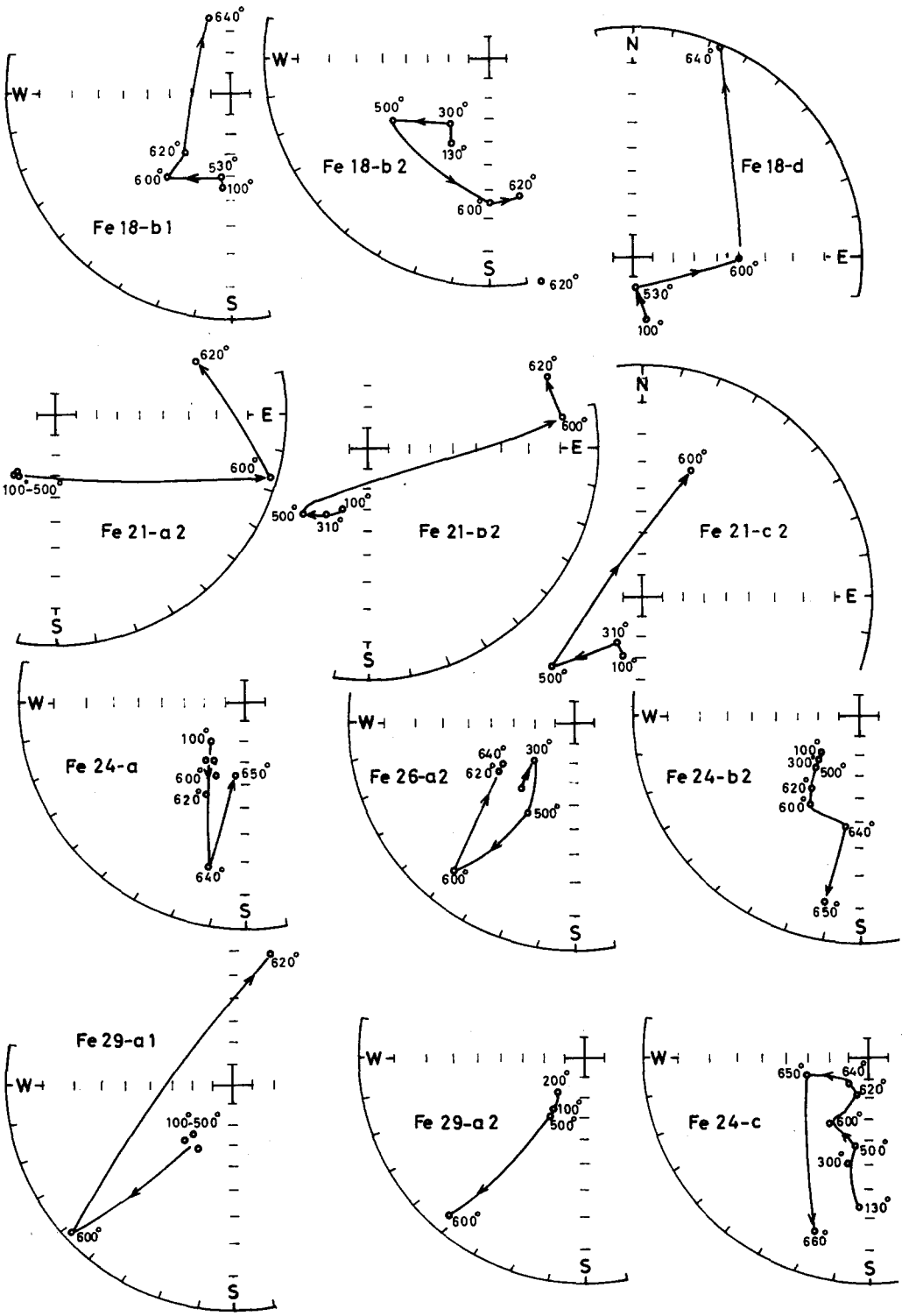


Fig. 4. Demagnetization results for carbonatite specimens. Other details as for Fig. 3.

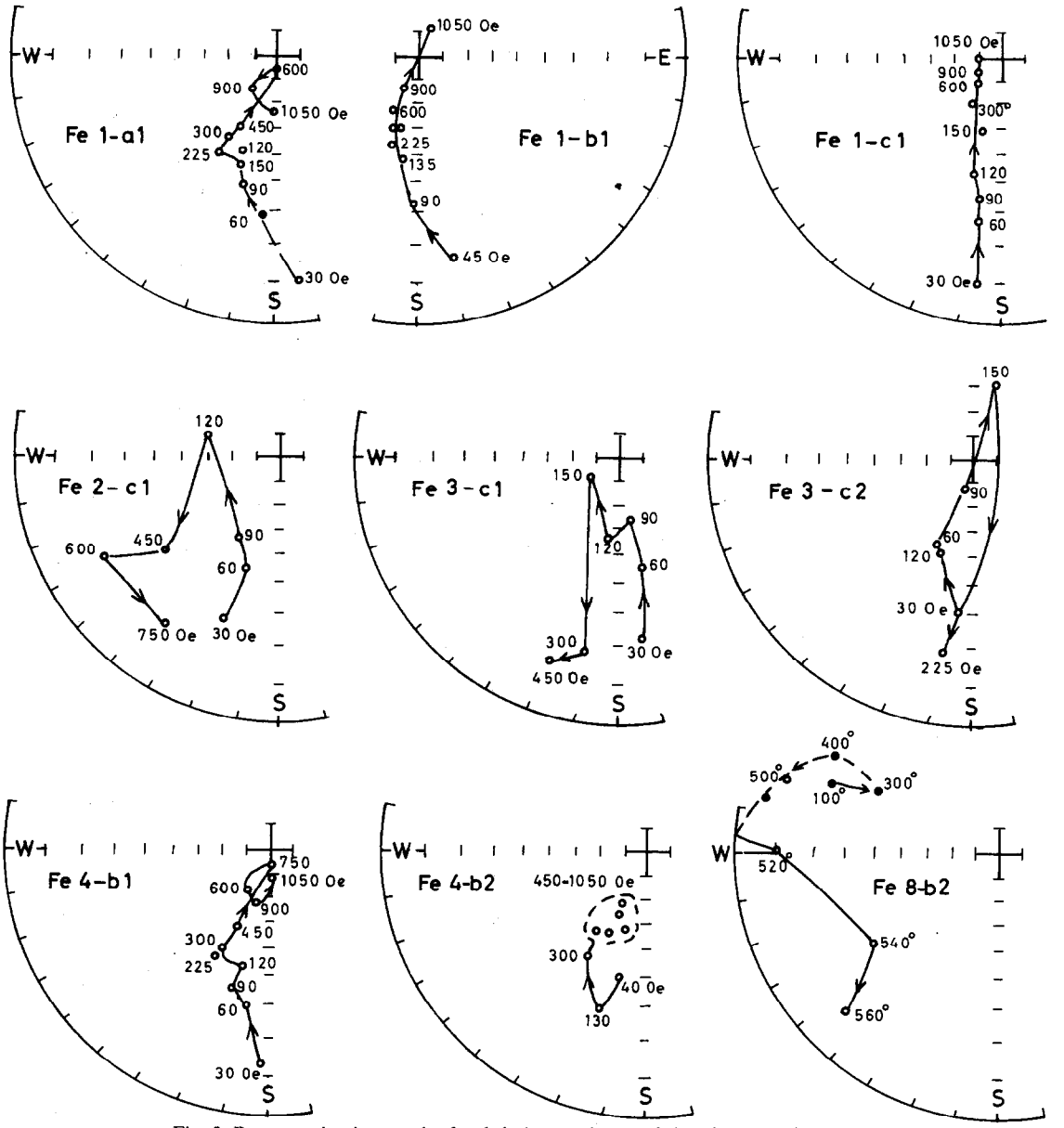


Fig. 5. Demagnetization results for dolerite specimens. Other details as for Fig. 3.

spinel phase which loses it magnetization between 500°C and 600°C. Thus, the temperature dependence of natural remanence as well as of saturation magnetization suggest at least two different magnetic substances in the carbonatite rock.

A number of representative examples of directional behaviour versus increasing demagnetization are shown by Fig. 3–5. There are critical stages in these analyses where well-defined directional variation, associated with a smooth intensity reduction, is superceded by erratic results, both in direction and intensity. The critical temperature or alternating field concerned may vary from specimen to specimen. The "erratic stage" (at a low intensity level) has obviously no geophysical relevance and the associated directions have therefore been ignored in the figures. There are sev-

eral plausible reasons for such experimental problems. Some measuring data suggest that the problem is in part caused by induced magnetization (possibly combined with an increase in unstable magnetism during demagnetization) becoming the dominant component after a major J_n-reduction. The observed susceptibility increase (by several per cent) in carbonatite specimens after heating above 600°C may have a direct connection with the meaningless results often encountered after demagnetization above 620°C or so. However, such odd observations may equally well be explained by an increased ability for acquiring T.R.M.'s when cooling the specimens from above 600°C in weak uncompensated fields. Whatever its cause, the experimental problems have effectively destroyed a satisfactory estimate of the stable magnetic sub-components concerned with in this study.

The basic requirement for acceptance of palaeomagnetic data is that there must be satisfactory experimental verification for either directional constancy over the entire stability range of the remanence or for any systematic direction change having terminated. The reason why this study is lacking numerical palaeomagnetic data is due to the fact that a group(s) of reliable "stable points" have not really been achieved. However, the remanence behaviour during demagnetization is fairly well defined and, therefore, the directional composition seems to be elucidated, at least to some extent. Particular demagnetization details for the different rock types can be summarized as follows.

Carbonatite. Owing to the high oxidation state of the ore minerals the magnetization of this rock has been analyzed by thermal demagnetization only. After heating to 100°C the magnetization direction is reverse with a moderate-steep upward inclination. However, on further demagnetization the fossil magnetism apparently splits into two nearly horizontal subcomponents of opposite polarity. A major problem in estimating the two directions is that they are both associated with similar blocking temperatures though on average the normal component has a slightly higher thermal stability than the reversed one. By judging from a large number of examples both polarity directions appear to have fairly shallow inclinations. Directional data for carbonatite specimens are shown in Fig. 4.

Kimberlite. Experimental data are shown in Fig. 3. Though the original magnetization is considerably more scattered than for the carbonatite the sub-components tend to be better defined. Thus, "the reversed component" is probably not far from that apparently established by specimens 11-a2 and 11-b. At least in specimen 11-a2, the most stable component seems to be normally directed and one does not know to what extent this latter component has a significant upsetting effect on the "stable" direction below approximately 500°C. The normal component is probably defined by the high temperature results (570°C and 600°C) for specimen 14-b1.

Dolerite. The two-component system suggested above is apparently present here as well but the normal component is less important than for the preceding rocks. In sample no. 4 specimen directions tend to establish a stable reversed and moderate-steep upward pointing magnetism vector after some initial alternating field treatment, i.e. in the original position of the carbonatite magnetization. However, when seen in conjunction with the results of specimens from other samples of the same body these "intermediate" results are thought to be false palaeomagnetic directions: being the accidental product of at least two sub-components of different polarity having identical stability spectra in the higher coercivity region. In this connection it is necessary to stress that a directional stability alone can not be considered as a (sufficient) measure for palaeomagnetic reliability unless the direction concerned is repeated in a larger number of specimens. Results are given in Fig. 5.

5. The suggested magnetization history

The haematite carbonatite is considered to be a metasomatic rock of probably perimagmatic origin (Saether, 1957). As suggested from inclusions and metasomatic transition zones this carbonatite rock must in part be the product of an assimilation of kimberlitic material. Hydrothermal metasomatic alterations effecting a formation over a somewhat longer time interval may drastically alter the initial magnetic structure (Storetvedt. 1970). For example, if mineralogical changes occur at times of field inversions (axis may be constant) deviating composite magnetizations of high stability may develop. There is some evidence that such a polarity change took place at about the time the Fen complex was formed though Permian

thermo-chemical effects (from the Oslo graben formation) may have complicated the picture.

The suggested normal magnetization component seems to be the predominating high-stability magnetization for the haematite carbonatite as well as the kimberlite. With reference to the foregoing chapter it is not possible to estimate this direction by numerical figures but there is reasonable evidence for suggesting a position at approximately 030 E, 10° up (negative). Such a direction would correspond to the suggested Lower–Middle Palaeozoic field for Western Europe (Briden et al., 1970; Storetvedt, 1967).As the Caledonian orogeny probably had negligible magnetization effects in southeast Norway, and no other potential remagnetization agency in the area prior to the Permian is known, the normal magnetization component is thought to date back to the formation of the Fen complex. However, the bulk magnetization is reverse so, it selfreversal mechanisms are not involved, an appreciable magnetization must have been acquired along one or more reversed directions. Fields who suggest themselves are firstly the one along the assumed primary axis (approximately 210, +10) and secondly the Permian one. It is now thought that the Permian palaeo-inclination deduced from rocks in the Oslo graben (Van Everdingen, 1960) is slightly overestimated due to Mesozoic remagnetization (Storetvedt and Petersen, 1970); the true palaeomagnetic field relative to southeast Norway being probably close to 200,–20 (Halvorsen, 1970, 1972). If the reversed magnetization was entirely imposed in Permian times one would expect to have a convergence of demagnetization trends towards this mean direction provided one considers specimens with an insignificant high stability normal component (it seems reasonable to suppose that any Permian thermal or thermo-chemical remagnetization would have lasted for a sufficient lenght of time to leave an approximate dipole field reading even in single specimens). However, a number of demagnetization trends move beyond the "Permian position" and attaining a practically horizontal direction before they either become dominated by the more stable normal component (cf. for instance specimen 29-a1 of Fig. 4) or the magnetic trends become disrupted in a random way. One is inclined to believe therefore that if a true reverse direction could be experimentally determined in these cases it would be along the supposed primary magnetization axis. Though the entire magnetization distribution in the Fen complex can be reasonably attributed to an "original" two-polarity magnetization (along the same axis) one can not rule out the possibility of a certain Permian contribution (reverse). For example, a Permian reheating may well have caused a partial reduction of the ferric oxides in the carbonatite producing the observed spinel phases. Such mineral changes are in accordance with recent experiments by Johnson (1972). Also, if a possible Permian component is of slightly lower stability than the older components present it would be impossible by conventional methods to extract it from the resultant. The suggestion of a Permian component is strengthened by studying the magnetic behaviour of the dolerites in comparison with the K/Ar ages obtained. From a palaeomagnetic point of view there is no reason for assuming a Permian age of the dolerites in that they show a similar remanence behaviour on demagnetization as do the two typical Fen rocks studied here. This does not necessarily imply that the dolerite dikes represent the final stage in the Fen magmatism but merely that the Lower–Middle Palaeozoic field is likely to be represented, i.e., the origin of the dolerites seems to be pre-mid-Carboniferous (cf. Storetvedt, 1967). The normal component of the dolerites is less pronounced than in the two other rock types studied. The explanation may well be that the metasomatism in the Fen area had ceased (entirely or nearly) at the time of dolerite injection so that the original mineralogical state may have remained relatively unaltered prior to a Permian activity in the area. The typical Fen rocks had already been through extensive mineralogical changes during their earliest history and providing therefore a mineralogy fairly resistant to further alterations while the dolerites may have been much more susceptible to further changes (for example in the Permian). This may explain why the alkaline/carbonatite rocks tend to give radiometric dates close to their true geologic age while the K/Ar chronometer of the dolerites probably became completely reset in the Permian.

6. The Lower Palaeozoic geomagnetic field for Western Europe

Before presenting the geophysical implication of

this study, the reliability of some palaeomagnetic data behind postulated deviating poles of the Palaeozoic will be discussed briefly. Unfortunately, it has become a tradition among palaeomagnetists to present their data in terms of mean directions, examples of detailed experimental results being very rare indeed. The picture is further distorted in that the important term "stable" apparently has a very wide meaning. Thus, in a recent study of Briden (1970) on the Upper Silurian Garabal Hill–Glen Fyne complex specimens from five sites were said to reach stable directional end-points after magnetic cleaning. However there is no confirmation of a final stage having been reached in any of the examples given (cf. his fig.5.). Also, the distribution of "stable directions" after cleaning is markedly meridional and suggesting therefore the presence of more than one relatively stable component. The results suggest that if careful demagnetization had been continued the terminal direction would define a nearly horizontal magnetization vector with north-northeasterly declination (cf. the direction trend of the thermally demagnetized specimen in fig. 5b of Briden after removal of the low-stability magnetization); in agreement with the axis predicted in the present study. Consequently, there seems to be no geomagnetic basis for the deviating pole proposed. Another"deviating" result (Deutsch and Somayajulu, 1970) has been proposed for some Ordovician strata (the Killary harbour ignimbrites of West Ireland). The sample mean directions which are thought to give the relevant field presentation (their fig. 3c) are strongly scattered (in the southwest quadrant). Furthermore, smeared within site distributions are a pronounced feature alongside a tendency of magnetization directions to perform northerly movement on demagnetization (cf. their fig. 2 and 3). There is no doubt that these ignimbrites are magnetically extremely complex and any conclusion seems premature at the present stage of analysis. A discussion of another divergent result from the British Isles suggesting similar remanence anomalies, has been given elsewhere (Storetvedt, 1971).

After the present paper was submitted for publication another study of the Fen complex magnetization has been published (Poorter, 1972). Based on results from the haematite carbonatite (rödberg) the author proposes another aberrant pole for the Palaeozoic of Europe. However, his final directions, which correspond closely to the initial rödberg directions of the present investigation, exhibit a clear meridional distribution with an inclination spread more than twice the declination variation. It is important to realize that this smeared distribution closely forms a segment of a great circle through the two sub-components suggested from the present analysis. Two "primary" antiparallel magnetizations, which in general define a resultant direction at variance with the underlying axis, are unlikely to have a constant relative contribution throughout the formation whereby at least a certain smeared distribution of the "stable" bulk magnetization (as seen in Poorters data) easily will develop. The largely oversimplified basic assumptions applied by Poorter is certainly the reason why the magnetization intricacy of the Fen complex apparently has escaped his attention completely.

In conclusion, there are good reasons for believing that the discrepant Palaeozoic palaeomagnetic data do not directly reflect the directions of ancient geomagnetic fields. Considering the radiometric dates of the Fen rocks as close estimates of the true age of the magmatic activity it seems that the Palaeozoic polar pattern relative to western Europe has a very simple structure; the only polar shift seems to occur around mid-Carboniferous (Storetvedt, 1967). Therefore, the relative "Lower Palaeozoic pole" thus defined, at around 160E 20N (in present grids), was apparently fixed for at least 300 m.y.

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